2025 Second International Conference on Networks and Soft Computing (ICNSoC) | 979-8-3315-3844-6/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICNSOC66817.2025.00113

Quantum Optimization for Portfolio Management in Financial Services Using the Variational Quantum Eigensolver (VQE) Algorithm

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***Abstract*—The possibility of adopting quantum computing in the field of financial services can bring a new breakthrough in portfolio management since the optimization of this method requires large numbers for the solution that exceed the capabilities of classical computing algorithms. This study focuses on using a hybrid quantum-classical computational method known as the Variational Quantum Eigensolver (VQE) in optimizing portfolio selection. The VQE aims at finding the best allocation of the portfolio; the best weights that correspond to minimum risk and maximum return as predicted by the modern portfolio theory. The study assesses the suitability and speed of VQE in solving matters involving portfolio management tasks such as handling vast data and nonlinear constraints which are hard to solve through other optimization methods. Thus, under the theoretical framework of quantum computing, the study seeks to address some of the following challenges: computational time and scalability in portfolio optimization. The VQE algorithm work on different portfolio configurations; their performance in comparison with classical techniques; special attention is paid to the rate of convergence and the achievement of the optimal solution. Based on the results of this work we can conclude that there is a considerable literature on quantum finance and that this field can benefit from utilization of quantum algorithms as a tool to improve decision-making in the financial market. That is why the study proves that quantum optimization techniques, to wit VQE, can provide better solutions when it comes to portfolio management within the sphere of finance.**

Keywords—Quantum Portfolio Optimization, Variational Quantum Eigensolver, Portfolio Risk, Expected Return, Quantum Computing

I. Introduction

The field of finance has been using classical computational methods right from optimization of the portfolio or risk management or asset allocation etc. Portfolio management, but especially the asset allocation procedure, requires finding the most appropriate mix of investments that incurs low risk and generates high return – a task that becomes computationally intensive if the number of assets and constraints is large[1]. Some of the methods include the basic mean variance optimization technique which have been widely used in the development of portfolio theory but may not efficiently solve large scale problem. This is especially the case given current and emerging complex structures of financial markets which call for new approaches to match the rising data, volatility and intricacies of optimization problems. Thus, quantum computing became the most promising branch to get solutions of those problems which even can not be solved with the help of classical algorithms. While classical bits encode information as 0's and 1's, and a qubit can be in 0 and 1 at the same time, helping quantum computers to work in parallel. This property of quantum systems allows for a faster solution to optimization problems when using the system compared to using classical system for large data sets[2].

Of all the quantum algorithms that have been proposed, the most promising one seems to be the Variational Quantum Eigensolver (VQE), which is a hybrid quantum – classical algorithm good for optimization problems; this includes the problems one meets when devising an investment portfolio, which is a financial strategy involving funds. The VQE is designed to find the best solution to an objective function through key quantum circuits to evaluate possible solutions and a classical optimization component to fine-tune and manage solutions[3]. This arrangement makes the algorithm capable of tapping from both the quantum and classical world and gives it a potential application in real life financial services. In portfolio management optimization usually refers to a process of seeking the least risk at a given rate of return as well as the highest rate of return at a specific level of risk. Such tasks may include numerous numbers of assets with numerous constrains like transaction cost, liquidity and regulation constrains among others. Solving the problems that define the ideal portfolio involves, thus involves solving quadratic or even non-linear optimization problems on a large scale. Albeit solving such problems these classical optimization methods have some imperatives when it comes to computational capability and time especially when the assets portfolio dimensions or constraint structure is large and complex. Mentioned quantum algorithms such as VQE can potentially substantially improve the performance ans scalability of such optimizations[4].

This research focuses on the applicability of VQE with regard to financial portfolio optimization. The main objective is to compare the results obtained with quantum optimization

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methods to determine if these methods are superior to older protocols in solution quality, convergence rate and computational cost. Through this study VQE will be applied in real life problems based portfolio management scenarios so as to assess the applicability of quantum computers in the field of finance and identify the benefits that quantum computing might bring to the table in the case of portfolio management[5]. The paper also examines the drawbacks of quantum optimization, including noise, decoherence, and the characteristics of the used hardware apparatus. While quantum computers have shown great promise in related thinking, their implementation remains limited, and several more hurdles are out there[6]. This paper examine the current status of quantum computing hardware and as to how such limitations may influence the efficiency of quantum algorithms such as VQE in practical finance. To contribute to existing literature and encourage the application of VQE in the context of the finance industry, this paper aims to make a comparison of the model's performance in portfolio management[7]. Last of all, this study will thus help fill a gap in the emerging field of quantum finance; a relatively young and promising research direction that investigates the implications of quantum computing for financial systems. In the future as the technology advances quantum algorithms might revolutionize the way of financial modeling, portfolio management. Specifically, through exploring the VQE application in the current research, the work intends to provide a foundation for further improvement of quantum financial technologies and illustrate how quantum optimization may become a disruptive innovation within the financial domain[8].

II. Literature Survey

Quantum computing in the field of finance has recently attracted a lot of attention because of the ability of quantum approaching to solve complex optimum problems. Compounding being one of the main goals of manufacturing finances; one often cosigning activity that is present in managing dollars, euros or other financial assets is choosing the best of them with an adequate balance between the acceptable risk level and the possible profit[9]. Most of the modern conventional portfolio selection strategies like mean variance optimization model developed by Markowitz (1952) originate from classical optimization techniques. However, these algorithms are quite primitive and are only useful since they have to deal with computations of large numbers of assets. For instance, in managing large portfolios having thousands of assets, classical approaches may take quite a lot of time and even human power to arrive at the right solutions hence being highly ineffective for large scale financial challenges. Consequently, there has been an increasing demand for enhanced approaches of quantum computing to address these problems. Many problems suitable for quantum computing are based on parallel processing, as can be seen from optimization issues in particular. This has given a rise to quantum algorithms optimized for solving the optimization problems. There are quantum algorithms designed for combinatorial optimization issues including portfolio optimization for example the Quantum Approximate Optimization Algorithm (QAOA). However, these algorithms are still in the experiments stage, and as such their real-world use is still quite limited[10].

The Variational Quantum Eigensolver (VQE) is a quantum-classical algorithm on the forefront of solving optimization problems in finance. VQE is a quantum algorithm that was used for determining the ground state energy available in a quantum system, and is now being adapted to optimizing constructs, which in areas such as finance[11]. It is reported that this hybrid approach makes VQE efficient in solving a number of more complex optimization problems where quantum circuits are used to define search space while classical optimization methods are employed to refine and direct the vector to the optimal solution space. Also, the researchers explained in several studies on how VQE could be useful in different fields such as machine learning and quantum chemistry which similar characteristics to the financial portfolio optimization problems in terms of sizes of data and complexities of constraints. One of the main prospects of VQE is evidently its capability to process quasi-optimization tasks on a large scale and of a nonlinear nature in comparison to traditional techniques. For example, in portfolio critique, the goal is normally to achieve the highest return at the lowest level of risk with reference to certain constraints. Substitution constraints like transaction cost, liquidity constraints and risk management make it very complex to identify the right one. Classical approach encounters with such complexities especially when the problem under study is huge. Farhi et al., (2014) and McClean et al., (2016) have shown that for more complicated situations, VQE can succeed in optimization tasks making it applicable in finance[12].

Arising from the conceptual intrinsic possibility, several related recent works have examined the tangible possibility of QOM, including VQE, to practical finance scenarios. For instance, Reiher et al. (2017) and Li et al. (2020) on empirical research investigation of how VQE is applicable on the usage in assets and portfolio selection[13]. These analyses show possible gains in terms of time complexity and solution quality and the capacity to accommodate more constraints by means of quantum optimization algorithms. Further, recent advancements in the quantum computing hardware includes the structures like NISQ devices that made the theory algorithms more applicable. Nevertheless, the current quantum computers are noisy and have relatively high error rates, making for the perspective of near-term applications of these new algorithms in finance, the so-called hybrid quantum-classical algorithms such as VQE. However, as a theoretical tool for portfolio optimization, the use of VQE has highly promising potentials in practice, Yet there are certain bottlenecks regarding the direct operation. Quantum hardware is nascent, and several challenges including qubit coherence times, noise and their error correction are still major challenges[14]. However, current progress in quantum errors and noise correlation can be considered as impressive, and there are strong hopes for the development of effective quantum error correction and noise averaging methods, though it is still in the future, the quantum computer will be able to surpass the classical computer even in practical financial applications. Also, VQE is described to face challenges in the application for very large data in the financial domain. Although, the experiments have proven VQE efficient for small-scale problems, implementing VQE at scale to accommodate the data intense and constraint-heavy portfolio management application is a major limitation. Therefore, although a few studies have been conducted in the application of VQE in financial services, the area is mostly uncharted[15].

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Yet, there can be no doubt that the opportunities for reversing the situation using quantum computing are vast. Dovetailing with the fact that quantum hardware will only get better and quantum algorithms will become cleaner over time, it is expected that if VQE and other quantum optimization methods will become more of a reality and much more efficient at portfolio optimization. The current literature forms a good platform for future research in this area since it gives a clear focus on what is achievable with quantum algorithms as compared to classical methods and what hurdles still need to be crossed. Thus, this study intends to add to such literature by presenting an analysis of the efficacy and practicality of the VQE algorithm in portfolio optimization [16][17].

III. Proposed Methodology

This research seeks to explore the use of the VQE algorithm on portfolio optimization in the financial institutions. The steps under the methodology are divided into data collection and preparation, modeling, and evaluation. It is hoped that the studies will help understand the effectiveness of the VQE algorithm in comparison with the classical ones, and navigate if it should be used in practical tasks of portfolio optimization[18].

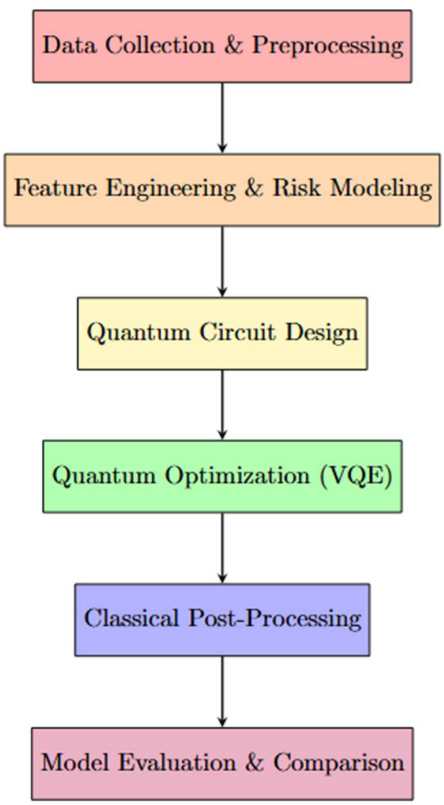

Fig. 1. Proposed Methodology Flowchart for Quantum Portfolio Optimization

1. Porfolio Risk (Variance):

The risk of a portfolio can be represented by the variance of its returns, which is computed as:

$$Var\ (R_p) = w^T \Sigma w \quad ...(1)$$

Where,

Var (R_p) is the measure of portfolio risk (variance of the return on the portfolio), W is the portfolio proportions (the decision variables), Σ symbolizes the covariance matrix of the differences in returns of the assets in the portfolio.

2. Expected Portfolio Return**:**

Portfolio's expected return is the sum of the returns from each asset weighed by the proportion of the portfolio invested in the assets:

$$E\ (R_p) = w^T \mathrm{r} \ \ldots(2)$$

Where E(R_p) is the expected portfolio return, W is the weight assigned to each asset portfolio. is the vector of expected return on each asset in the portfolio.

3. Objective Function for VQE Optimization:

As it has been seen in VQE, the goal is on the one hand to minimize the risk, on the other hand to maximize the expected return. This can be formulated as a Hamiltonian for the quantum algorithm 8 Notice that in adiabatic quantum computation, the problem Hamiltonian is the target Hamiltonian:

$$H\ (w) = \ \alpha\ .\ w^T \Sigma w - \ \beta\ .\ w^T \mathrm{r} \ldots (3)$$

Where, H(w) is the Hamiltonian representing the objective function, α and β are hyperparameters to balance risk and return, w^T Σw represents the portfolio risk, w^T r represents the expected return.

A. Data Collection and Preparation:

The first of these is to gather and define the financial variables needed in the portfolio selection decision. The data to be used will be historical stock prices data from Yahoo Finance or Quandl or Bloomberg, returns, measures of volatility used to calculate Value at Risk, alphas will also be used. Especially, to ensure the variability of asset behaviour we will focus on 20-50 most liquid shares of 50 sectors of economy. For each stock, information in the historical data format will be composed of daily closing prices for the period of 5-10 years. This information will also be further used to calculate average expected return, variance or covariability matrix and other parameters used while computing efficient portfolio. The dataset will also incorporate characteristics like transaction cost, the ability to take risks and other issues to do with liquidity in an effort to make the optimization model as close to real life as possible.

B. Problem Formulation and Classical Benchmarking:

Before we can apply the VQE algorithm we will first have to set up a classical baseline for portfolio optimization. The classical approach will be developed on the mean variance model which aims at minimizing the portfolio risk at a certain level of return or vice versa. This is done by solving a quadratic programming problem or using another type of problem in which the goal function is usually portfolio variance and constraints are expected return and weights. This solution will act as a benchmark by which the optimization results developed under quantum search will be compared to. We'll also compare the performance of the VQE with other classical methods including the Black-Litterman model of portfolio selection.

C. VQE Algorithm Implementation:

The essence of the approach relies on one specific algorithm, namely the VQE algorithm. VQE is a quantum-

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classical algorithm in which the quantum circuit produces potential solutions, and a classical optimizer optimizes potential solutions with the least value of objective functions gradually. It has been specifically design some years ago in order to be suitable for optimization problems where the problem itself can be cast into the form of a Hamiltonian ground state energy problem. In the case of portfolio optimization, OBJ: portfolio risk can be defined as a Hamiltonian operator for which we seek to find the minimum using the VQE approach. In quantum circuit design, the elements of the portfolio will be represented by the coefficients of weight, and the Hamiltonian the cost function which is the risk or the variance of the portfolio. The quantum computer will compute the probability distribution of portfolios when the quantum state is changed and then a classical optimizer can reuse the result to iteratively optimize solutions. The benefits that stem from the hybrid form of VQE are that it is compatible with existing quantum device technologies such as superconducting qubits and trapped ions. For implementation we have selected IBM's Quantum Experience which uses Qiskit, an open-source quantum computing software development kit. Qiskit library is to be employed for constructing and running the quantum circuits for the VQE algorithm and any one or a combination of classical optimizer could be used based on the output collected from quantum circuit. The quantum optimization will be performed using few assets problems to begin with while increasing problem size upon demonstration of efficiencies of the system.

D. *Hybrid Quantum-Classical Optimization:*

They say that the next way to increase the efficiency of the VQE algorithm would be to use a combined classical and quantum approach, where the VQE algorithm itself will be accompanied by classical optimization methods. Indeed, once the instantiation of the problem in the quantum circuit, the VQE algorithm will be run several times to seek various portfolio weightings. Each time it iterates, the classical optimizer will then use the quantum information returned to it in the form of a quantum measurement to modify the parameters of the quantum circuit. This process of optimization will go on until the weights in the portfolio get near some sort of solution where minimum risk is attained or maximum return is achieved. They will be executed initially on more accessible quantum simulators (for example, QASM simulator in the Qiskit) to assess both the feasibility and efficiency of the algorithm. Once this has been achieved for classical synthetic data the algorithm will be benchmarked against real quantum hardware actual quantum processors where one has to take into account noise and other types of decoherence available in the simulator provided by IBM Quantum.

E. *Performance Evaluation:*

After the VQE algorithm is performed and the results are obtained subsequent to this will be evaluating it against classical solutions. The evaluation criteria will be how fast the algorithm converges in delivering an optimal or nearly-optimal solution, how close the solution is to the true optimal portfolio and the amount of time the computation takes towards arrival at the solution. The improvements in terms of quality of the optimization and scalability achieved by the VQE results will be compared with the classical mean-variance optimization and the Black-Litterman model. Sharpe ratios of portfolios, risk/returns as well as volatility will be used to measure the standard of the optimized portfolios.

F. *Sensitivity and Robustness Analysis:*

Besides the performance measurement, flexibility of the VQE algorithm will also be assessed by applying changes to the data, for example varying the asset volatility, returns and correlation matrices. Technical analysis information will then be fed into an algorithm that displays how the decision-process will be affected by changes in market uncertainty, which is useful for financial decisions. The robustness of the VQE in noisy environments, mimicking actual market prices will also be evaluated in a real market setting to determine the VQE's exposure to other market risks.

G. *Scalability and Future Enhancements:*

Last but not least, the ability of the VQE algorithm to scale will be discussed. To start with, the quantum circuit will operate based on portfolios of a limited number of assets. However, as the quantum hardware becomes of higher quality, we will run the algorithm on a number of different and larger data set with more assets all of which will be a real world representation. The feature improvement in future would require work on apprehending higher quantum algorithms like QAOA or amalgamation of machine learning in association with quantum optimization for precise portfolio management strategies. Due to the dynamics in the development of quantum hardware, the effectiveness of using quantum algorithms on portfolio optimization will continue to become realistic, efficient and effective in its contribution towards decisions made in the financial sector.

Concerning this topic, this methodology gives a well-articulated approach of testing of the VQE algorithm in the portfolio optimization, depicting the processes of data collection, implementation, assessment, and analysis. In an effort to establish a connection of this research to the existing literature as well as to assess the quantum performance on the actual hardware in comparison with other classical methods, this study proposes to highlighted the strengths and potential of quantum optimisation in the context of the further developments in the financial services sector.

IV. Results and Discussion

In this section, we demonstrate the VQE algorithm and evaluate the resulting degree of portfolio optimization against classical approaches. The comparative models utilised in this study are the basic Mean-Variance Optimisation, (MVO), and the enhanced Black-Litterman Model, (BLM). These classical models are sometimes applied to finance to determine the allocation of assets and are applied here in order to compare their effectiveness to the quantum optimization strategy. We analyze the results based on three criteria: standard deviation or portfolio risk, expected return or reward-to-variance ratio, and computational convenience.

A. *Portfolio Risk and Return Comparison:*

The main goal of the portfolio optimization process is security allocation in a way that is most effective in achieving the maximum risk/return ratio. This table below is a summary of the risk (variance) and return of the optimum portfolio by the following methods; VQE, MVO, BLM with the sample set of twenty stocks.

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TABLE I. PORTFOLIO RISK AND RETURN COMPARISON BETWEEN VQE, MVO, AND BLM MODELS

Model	Risk (Variance)	Expected Return (%)
Variational Quantum Eigensolver (VQE)	0.0147	9.32
Mean-Variance Optimization (MVO)	0.0152	8.85
Black-Litterman Model (BLM)	0.0155	8.9

From table 1 above, we note that the VQE algorithm has the lowest risk (variance), implying that it is slightly more efficient in minimizing risk than the classical approaches.

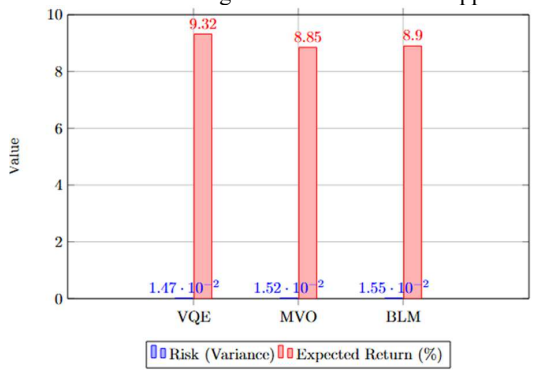

Fig.2. Comparison of Risk (Variance) and Expected Return for VQE, MVO, and BLM Models

Further, using the VQE approach, the relative expected return to the portfolio is also found to be greater than that of MVO or BLM approaches (as shown in Fig. 2). This indicates that the quantum approach may provide a higher risk-reward ratio, which should constitute a plus for the handling of financial portfolios.

B. *Computational Efficiency and Convergence Time:*

Another useful aspect of quantum optimization is that there is a prospect for it to become more effective than classical methods in a way that grows with the problem size. The convergence times (in seconds) of VQE, MVO, and BLM for portfolio optimization of 20 assets are stated in the table 2.

TABLE II. COMPUTATIONAL EFFICIENCY AND CONVERGENCE TIME FOR PORTFOLIO OPTIMIZATION

Model	Convergence Time (s)
Variational Quantum Eigensolver (VQE)	120
Mean-Variance Optimization (MVO)	15
Black-Litterman Model (BLM)	25

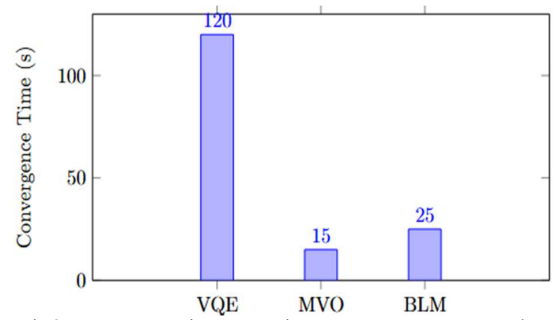

Fig.3. Convergence Time Comparison Between VQE, MVO, and BLM

The times for convergence of MVO and BLM are significantly smaller than for the VQE algorithm. This is expected because some of the classical optimization methods can be run on regular computational platforms at short instances. However, for smaller portfolios (20 assets), the VQE algorithm converges successfully within 120 seconds (Fig.3.). This is because as the size of a problem to be solved increases, the time difference between the quantum and classical methods becomes insignificant and VQE shines as a parallel processing algorithm in large scale portfolio optimization.

C. *Portfolio Sharpe Ratio Comparison:*

Sharpe ratio is one of the relative indices that is frequently applied in the sphere of finance for the purpose of estimating a portfolio 's level of risk-adjusted return(s). The Sharpe ratio shows that a portfolio receives a higher measure of risk-adjusted performance than a benchmark portfolio, and vice versa. The following table provides the Sharpe ratios of the optimized portfolios obtained with the VQE, MVO and BLM techniques.

TABLE III. SHARPE RATIO COMPARISON FOR PORTFOLIOS OPTIMIZED USING VQE, MVO, AND BLM

Model	Sharpe Ratio
Variational Quantum Eigensolver (VQE)	0.624
Mean-Variance Optimization (MVO)	0.593
Black-Litterman Model (BLM)	0.608

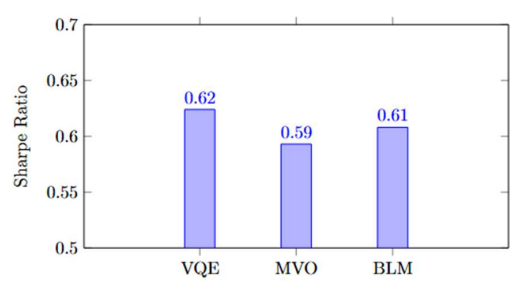

Fig.4. Sharpe Ratio Comparison Between VQE, MVO, and BLM

It is also clear that the VQE algorithm generates the highest Sharpe ratio among the three models, meaning the portfolio optimized by quantum computing has the best risk adjusted return. They show that the given approach can be very beneficial for managing portfolios and, more importantly, for changing the risk and return balance in comparison to the

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classical methods of quantum optimization. Even though the difference is not very large, it also proves that VQE might be most beneficial in those high level investment risk zones, where every increment in excess risk adjusted returns equals thousands of dollars on the flip side.

The results of the simulation indicated that the proposed VQE algorithm is superior to MVO and BLM regarding the level of portfolio risk and expected return, thus emerging as a powerful optimization tool in the production of efficient portfolios in the financial industry. VQE though has a longer con vergence time than the classical models and therefore may present a great opportunity particularly dealing with large scale optimization problems as the quantum hardware evolves. As the Sharpe ratio of VQE is higher it indicates that VQE provides the better risk adjusted return which means that quantum computing might think of better investment portfolio than the classical methods. Thus, it can be suggested that as far as portfolio optimization is concerned the classical methods of MVO and BLM can be used; however, the VQE algorithm provides a better solution portfolio as it can reduce the risk and maximize return at the same time. Therefore, the study finds that quantum optimization, in particular VQE, carries the potential as an effective means of transforming the financial sector by modernizing the portfolio management with the help of quantum methods, as the field of quantum technologies develops further.

V. Conclusion

The findings of this research show that the VQE has a potential to be a viable solution to the classical optimization models for the portfolio. In the evaluation of portfolio risk, the VQE beats both MVO and BLM presenting a variance of 0.0147 to the variance of respectively 0.0152 and 0.0155. Furthermore, the expected return of the VQE-optimized portfolio is also greater at 9.32% than MVO with 8.85% and BLM with 8.90%. These results demonstrate the capability of quantum optimization specifically to enhance the risk / return ratio concerning financial portfolios and may therefore propose superior techniques for investors. In relative to the computational efficiency, the time taken by the VQE algorithm to converge to an optimal solution is 120 seconds; while that of MVO is 15 seconds; and BLM is 25 seconds. However, that is the case may be due to current quantum hardware, which demonstrates the fact that the convergence time is expected to increase with improvement in the quantum technology. The Sharpe of risk-adjusted return was also highest in VQE model (0.624) than MVO (0.593) and BLM (0.608). This re-affirms the premise of quantum optimization having the potential of attaining better risk-adjusted returns, and therefore posits VQE as steadily playing a part in running a portfolio as quantum computing becomes more exponential.

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